SELF LEARNING ASSIGNMENT

DAY_01

04.10.2026

Q.1. Vasya has recently learned to type and log on to the Internet. He immediately entered a chat room and decided to say hello to everybody. Vasya typed the word s. It is considered that Vasya managed to say hello if several letters can be deleted from the typed word so that it resulted in the word "hello".

For example, if Vasya types the word "ahhellllloou", it will be considered that he said hello, and if he types "hlelo", it will be considered that Vasyagot misunderstood and he didn't manage to say hello. Determine whether Vasya managed to say hello by the given word s.

Input :
The first and only line contains the word s, which Vasya typed. This word consists of small Latin letters, its length is no less that 1 and no more than 100 letters.

Output :
If Vasya managed to say hello, print "YES", otherwise print "NO".

Examples:
Input : ahhellllloou
Output : YES
Input : hlelo
Output : NO

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DAY_02

05.10.2026

Q.2. Given an array of integers arr[], move all the zeros to the end of the array while maintaining the relative order of all non-zero elements.

Examples:

- Input: arr[] = [1, 2, 0, 4, 3, 0, 5, 0]

  Output: [1, 2, 4, 3, 5, 0, 0, 0]

  Explanation: There are three 0s that are moved to the end.

- Input: arr[] = [10, 20, 30]

  Output: [10, 20, 30]

  Explanation: No change in array as there are no 0s.

- Input: arr[] = [0, 0]

  Output: [0, 0]

  Explanation: No change in array as there are all 0s.

Q.3. Given a string s, the task is to check if it is palindrome or not.

Example:

- Input: s = "abba"
  Output: 1
  Explanation: s is a palindrome

- Input: s = "abc"
  Output: 0
  Explanation: s is not a palindrome

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DAY_03

06.10.2026

1. **Third largest element in an array of distinct elements**

Given an array of n integers, the task is to find the third largest element. All the elements in the array are distinct integers.

Examples:

- Input: arr[] = {1, 14, 2, 16, 10, 20}

  Output: 14

Explanation: Largest element is 20, second largest element is 16 and third largest element is 14

- Input: arr[] = {19, -10, 20, 14, 2, 16, 10}

  Output: 16

Explanation: Largest element is 20, second largest element is 19 and third largest element is 16

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2. **Reverse words in a string**

Given a string, find a way to reverse the order of the words in the given string.Note: string may contain leading or trailing dots(.) or multiple trailing dots(.) between two words. The returned string should only have a single dot(.) separating the words.

Examples:

- Input: s = "i.like.this.program.very.much"

  Output: much.very.program.this.like.i

Explanation: The words in the input string are reversed while maintaining the dots as separators, resulting in "much.very.program.this.like.i".

- Input: s = ”..geeks..for.geeks.”

  Output: geeks.for.geeks

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3. Remove duplicates from Sorted Array

Given a sorted array arr[] of size n, the goal is to rearrange the array so that all distinct elements appear at the beginning in sorted order. Additionally, return the length of this distinct sorted subarray.

Note: The elements after the distinct ones can be in any order and hold any value, as they don't affect the result.

Examples:

- Input: arr[] = [1, 2, 2, 3, 4, 4, 4, 5, 5]

  Output: [1, 2, 3, 4, 5

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4. Reverse a String –

Given a string s, the task is to reverse the string. Reversing a string means rearranging the characters such that the first character becomes the last, the second character becomes second last and so on.

Examples:

Input: s = "GeeksforGeeks"

Output: "skeeGrofskeeG"

Explanation: The first character G moves to last position, the second character e moves to the second-last and so on.

Input: s = "abdcfe"

Output: "efcdba"

Explanation: The first character a moves to last position, the second character b moves to second-last and so on.

DAY_04

07.10.2026

4. Suneet and Slavic play a card game. The rules of the game are as follows

- Each card has an integer value between 1 and 10.
- Each player receives 2 cards, which are face-down (so a player doesn't know their cards).
- The game is turn-based and consists exactly of two turns.
- In a round, both players pick a random unflipped card and flip it.
- The player who flipped a card with a strictly greater number wins the round.
- In case of equality, no one wins the round.
- A player wins the game if he wins the most number of rounds (i.e., strictly more than the

other player). In case of equality, no one wins the game.

Since Suneet and Slavic aren't best friends, you need to calculate the number of ways the game

could happen that Suneet would end up as the winner.

Input

The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases.

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Each of the next t lines contains 4 integers a1, a2, b1, b2 (1 ≤ a1, a2, b1, b2 ≤ 10) where:

- a1 and a2 are the two cards Suneet has,
- b1 and b2 are the two cards Slavic has.

Output

For each test case, output a single integer — the number of games Suneet would win considering all

possible ways the game could happen.

Example Input

5

3 8 2 6

1 1 1 1

10 10 2 2

1 1 10 10

3 8 7 2

Example Output

2

0

4

0

2

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DAY_05

08.10.2026

3. You have a stripe of checkered paper of length n. Each cell is either white ('W') or black ('B').

Your task is to find the minimum number of cells that must be recolored from white to black

in

order to have a segment of k consecutive black cells on the stripe.

If a segment of k consecutive black cells already exists, you should print 0.

Input

- The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases.
- The descriptions of t test cases follow.
- Each test case contains:
  - A line with two integers n and k ($1 \le k \le n \le 2 \cdot 10^5$).
  - A line of length n, consisting of letters 'W' (white) and 'B' (black), representing the

stripe.

It is guaranteed that the sum of all n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print a single integer — the minimum number of white cells ('W') that need to

be recolored to black in order to have a segment of k consecutive 'B' characters.

Example Input

4

5 3

BBWBW

5 5

BBWBW

5 1

BBWBW

1 1

W

Example Output

1

2

0

1

Explanation

● Test case 1: "BBWBW" — You can flip the 'W' at position 3 to 'B' → "BBBBW" → segment of length

3 of 'B' → only 1 repaint needed.

● Test case 2: "BBWBW" — Flip positions 3 and 5 → "BBBBB" → now a full segment of 5 'B'

→ 2 repaints.

● Test case 3: Already has at least one 'B', and k=1 → no repaints needed.

● Test case 4: "W" → must repaint it to 'B' → 1 repaint.

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4. You are given list of n integers. You can perform the following operations

● Choose an element x from the list.

● Erase x from the list.

● Subtract x from all remaining elements in the list.

This operation reduces the list size by exactly 1.

You are also given a target value k. Your task is to determine whether it is possible to perform

exactly n-1 operations so that only one number remains in the list and it is equal to k.

Input

● The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases.

● The description of t test cases follows.

- For each test case:

○ The first line contains two integers n and k ($2 \le n \le 2 \cdot 10^5$, $1 \le k \le 10^9$) — the number

of elements in the list and the target value.

○ The second line contains n integers a_1, a_2, ..., a ($-10^9 \le a_i \le 10^9$) — the initial

list.

It is guaranteed that the sum of all n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print "YES" if you can perform the operations to make the final remaining

number equal to k. Otherwise, print "NO".

You may print the result in any letter case.

Example Input

4

4 5

4 2 2 7

5 4

1 9 1 3 4

2 17

17

0

2 17

18

18

Example Output

YES NO YES

NO

Explanation

Test Case 1:

Initial: [4, 2, 2, 7], Target: 5

One possible sequence:

- Remove 2 → [4, 2, 7] becomes [2, 0, 5]
- Remove 2 → [0, 5] becomes [−2, 3]
- Remove −2 → [3] becomes [5]

So, output is YES.

Test Case 2:

[1, 9, 1, 3, 4] → No sequence of operations leads to 4 → NO.

Test Case 3:

[17, 0] → remove 0 → [17] YES

Test Case 4:

[18, 18] → whatever you remove, the remaining becomes 0 → not 17 → NO.

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5. Alice and Bob are playing a game. They have n cards numbered from 1 to n. At the beginning of the game, some of these cards are given to Alice, and the rest are given to Bob. Card with number i beats card with number j if and only if i > j, with one exception: card 1 beats card n.The game continues as long as each player has at least one card. During each turn, the following occurs:

- Alice chooses one of her cards and places it face up on the table;
- Bob, seeing Alice's card, chooses one of his cards and places it face up on the table;
- if Alice's card beats Bob's card, both cards are taken by Alice. Otherwise, both cards are taken by Bob.

A player can use a card that they have taken during one of the previous turns.

The player who has no cards at the beginning of a turn loses. Determine who will win if both players play optimally.

Input

The first line contains a single integer t (1 ≤ t ≤ 5000) — the number of test cases.

Each test case consists of two lines:

- The first line contains a single integer n (2 ≤ n ≤ 50) — the number of cards;
- The second line contains n characters, each either A or B. If the i-th character is A, then card number i is initially given to Alice; otherwise, it is given to Bob.

Additional constraint on the input: in each test case, at least one card is initially given to Alice, and at least one card is initially given to Bob.

Output

For each test case, output Alice if Alice wins with optimal play, or Bob if Bob wins. It can be shown that if both players play optimally, the game will definitely end in a finite number of turns with one of the players winning.

Example

Input

8

2

AB

2

BA

4

ABAB

4

BABA

3

BAA

5

AAAAB

5

BAAAB

6

BBBAAA

Output

Alice Bob

Bob Bob

Alice

Alice Bob

Alice

Note

In the first test case, Alice has only one card, and Bob has only one card. Since Alice's card beats

Bob's card, she wins after the first turn.

In the second test case, Alice has only one card, and Bob has only one card. Since Bob's card beats

Alice's card, he wins after the first turn.

In the third test case, there are two possible game scenarios:

- if Alice plays the card 1 on the first turn, Bob can respond with the card 2 and take both cards. Then, Alice has to play the card 3 on the second turn, and Bob will respond by playing the card 4. Then, he wins;

- if Alice plays the card 3 on the first turn, Bob can respond with the card 4 and take both cards. Then, Alice has to play the card 1, and Bob can respond either with the card 2 or the card 3. Then, he wins. In the fourth test case, there are two possible game scenarios:

- if Alice plays the card 2 on the first turn, Bob can respond with the card 3 and take both cards. Then, Alice has to play the card 4 on the second turn, and Bob will respond by playing the card 1. Then, he wins;

- if Alice plays the card 4 on the first turn, Bob can respond with the card 1 and take both cards. Then, Alice has to play the card 2, and Bob can respond either with the card 3 or the card 4. Then, he wins.

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DAY_06

09.10.2026

6. On another boring day, Egor got bored and decided to do something. But since he has no friends, he came up with a game to play.

Egor has a deck of n cards, the i-th card from the top has a number a[i] written on it. Egor wants to play a certain number of rounds until the cards run out. In each round, he takes a non-zero number of cards from the top of the deck and finishes the round. If the sum of the numbers on the cards collected during the round is between l and r, inclusive, the round is won; otherwise, it is lost.

Egor knows by heart the order of the cards. Help Egor determine the maximum number of rounds he can win in such a game. Note that Egor is not required to win rounds consecutively.

Input

Each test consists of several test cases.

The first line contains an integer t ($1 \le t \le 10^4$) — the number of test cases. This is followed by a

description of the test cases.

The first line of each test case contains three integers n, l, and r ($1 \le n \le 10^5$, $1 \le l \le r \le 10^9$).

The second line of each test case contains n integers $a_1, a_2, \ldots, a$ ($1 \le a_i \le 10^9$) — the numbers

on the cards from top to bottom.

It is guaranteed that the sum of n for all test cases does not exceed 2×10^5.

Output

For each test case, output a single number — the maximum number of rounds Egor can win.

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Example

Input

8

5 3 10

2 1 11 3 7

10 1 5

17 8 12 11 7 11 21 13 10 8

3 4 5

3 4 2

8 12 25

10 7 5 13 8 9 12 7

2 3 3

5

2

9

7 9

2

10 5 1 3 7 6 2

3

1

8 10

9

5

5 6

1

4 2 6 4

Output

3

0

1

4

0

3

1

2

Note

In the first test case, Egor can win 3 rounds:

● In the first round, take the top 2 cards with values 2 and 1 and win, as their sum is 3. After

this, the deck will look like this: [11, 3, 7].

● In the second round, take the top card and lose, as its value 11 is greater than r = 10.

After this, the deck will look like this: [3, 7].

● In the third round, take the top card with value 3 and win. After this, the deck will look like this: [7].

● After this, in the fourth round, Egor only has to take the last card in the deck with value 7 and win again.

In the second test case, Egor cannot win any rounds, no matter how hard he tries.

In the third test case, you can take one card in each round, then the first and third rounds will be losing, and the second round will be winning.

In the fourth test case, you can take two cards in each round and always win.

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7. Polycarp has just finished writing down the lecture on elvish languages. The language this week was "VwV" (pronounced as "uwu"). The writing system of this language consists of only two lowercase Latin letters: 'v' and 'w'

Unfortunately, Polycarp has written all the lecture in cursive and without any spaces, so the notes look like a neverending sequence of squiggles. To be exact, Polycarp can't tell 'w' apart from 'vv' in his notes as both of them consist of the same two squiggles.

Luckily, his brother Monocarp has better writing habits, so Polycarp managed to take his notes and now wants to make his own notes more readable. To do that he can follow the notes of Monocarp and underline some letters in his own notes in such a way that there is no more ambiguity. If he underlines a 'v', then it can't be mistaken for a part of some 'w', and if the underlines a 'w', then it can't be mistaken for two adjacent letters 'v'.

What is the minimum number of letters Polycarp should underline to make his notes unambiguous?

Input

The first line contains a single integer t ($1 \le t \le 100$) — the number of testcases.

Each of the next t lines contains a non-empty string in VwV language, which consists only of

lowercase Latin letters 'v' and 'w'. The length of the string does not exceed 100.

Output

For each testcase print a single integer: the minimum number of letters Polycarp should underline so

that there is no ambiguity in his notes.

Example

Input

5 vv v

w vwv

vwvvwv

Output

1

0

1

1

3

Note

In the first testcase it's enough to underline any of the two letters 'v'.

In the second testcase the letter 'v' is not ambiguous by itself already, so you don't have to underline

anything.

In the third testcase you have to underline 'w', so that you don't mix it up with two letters 'v'. In the

fourth testcase you can underline 'w' to avoid all the ambiguity.

In the fifth testcase you can underline both letters 'w' and any of the two letters 'v' between them.